

SOFR • FX • XCCY • QUANT SERIES

SOFR, FX & Cross-Currency Curves

US dollar rates, FX and cross-currency
curves with QuantLib Python



Rates with Bip

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SOFR, FX & Cross-Currency Curves. US dollar rates, FX and cross-currency curves with QuantLib Python.

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The SOFR swap, SOFR futures, FX swap, cross-currency basis and AUD swap quotes in this book are illustrative, made up for teaching. They are not market data.

Chapters 1 and 2 use SOFR, EFR and the SOFR Averages and Index published by the Federal Reserve Bank of New York. The SOFR and EFR data are subject to the Terms of Use posted at newyorkfed.org. The New York Fed is not responsible for publication of the SOFR and EFR data by Rates with Bip, does not sanction or endorse any particular republication, and has no liability for your use. SOFR data are calculated using data provided under a license granted to the New York Fed by DTCC Solutions LLC; Solutions, its affiliates, and third parties from which they obtained data have no liability for the content of this material. Rates with Bip is not affiliated with the New York Fed. The New York Fed does not sanction, endorse, or recommend any products or services offered by Rates with Bip.

The AUD/USD exchange rates in Chapter 4 are the noon buying rates in New York from the Federal Reserve Board's H.10 release. Source: Board of Governors of the Federal Reserve System.

Market conventions are checked against the sources listed at the end of each chapter. Conventions change; always confirm against the current documents of the New York Fed, the ARRC, CME, AFMA, the RBA and your clearing house before relying on them.

Every number in the text, tables and figures is computed by the companion QuantLib Python notebooks (QuantLib 1.43), one per chapter, which reproduce each result.

This book is for education only. It is not financial, investment, legal or tax advice.

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How to use this book

This book follows the ten video episodes of the course, one chapter per episode, and goes a little deeper in each. It is the sequel to *AUD Swaps & Curves*. That book built Australian dollar curves from AONIA and BBSW; this one crosses the Pacific to US dollar rates, the AUD/USD exchange rate, and the cross-currency swaps that tie the two together. It is written for people on rates, FX and treasury desks, in market risk and product control, and for quants who know the maths but not the market conventions.

The path runs from the US dollar overnight rate to an Australian dollar curve for trades collateralised in US dollars:

- **Chapters 1–3** cover US dollar rates: the Fed's target range, EFFR and SOFR; exactly how a SOFR swap pays, using real fixings from the New York Fed; and the SOFR curve, built from swaps and futures.
- **Chapters 4–5** turn to foreign exchange: AUD/USD spot and the spot date, with real exchange rates from the Federal Reserve Board, then FX forwards and FX swaps, covered interest parity, and the cross-currency basis.
- **Chapters 6–8** put the two together: BBSW/SOFR cross-currency basis swaps, why the collateral currency decides the discount curve, and the Australian dollar curve for US dollar collateral, bootstrapped from FX swaps and basis swaps.
- **Chapters 9–10** use that curve: long-dated FX forwards, an off-market basis swap, and finally risk and hedging.

What you need from the first book

The AONIA and BBSW curves of *AUD Swaps & Curves* appear here as ready-made inputs from Chapter 5 on. This book says what each one is when it first needs it, but it doesn't derive them again. If a step with AONIA or BBSW goes by too fast, Chapters 3 to 6 of the first book build those curves from the start.

Numbers you can reproduce

Every number in the text, tables and figures is computed by a companion QuantLib Python notebook, one per chapter. Nothing is typed in by hand. Where a chapter works an example by hand, it also shows that QuantLib gets the same answer. You are encouraged to run the notebooks, change the inputs and watch the results move.

The notebooks are free to download from github.com/RatesWithBip/curriculum, in the `sofr-fx-and-cross-currency-curves` folder, and each has an *Open in Colab* button. They run as they are in Google Colab: each installs QuantLib and writes its own data, so there is nothing else to download.

About the data

Two kinds of data appear in this book.

- **Real data.** Chapters 1 and 2 use SOFR, EFFR and the SOFR Index as published by the Federal Reserve Bank of New York. Chapter 4 uses AUD/USD exchange rates from the Federal Reserve Board's H.10 release.
- **Illustrative quotes.** From Chapter 3 on, the market is made up for teaching: SOFR swap rates and futures prices, FX forward points, basis swap spreads and Australian dollar swap rates. They are shaped like a plausible market, with the signs and rough sizes of the real one, but they are not market data. Dealer quotes are licensed data, and the ideas don't depend on the exact levels.

About the conventions

Every convention stated in this book is checked against a primary source: the New York Fed and the Alternative Reference Rates Committee for SOFR, CME's rules as filed with the CFTC for SOFR futures, AFMA's *Interest Rate Derivative Conventions* and its BBSW/SOFR cross-currency term sheet, and the Reserve Bank of Australia. The sources are listed at the end of each chapter. Where the model in this book departs from a term sheet, as it does once in Chapter 8, the chapter says so and measures what it costs.

Conventions do change. Before relying on one in practice, check the current version of the source, and your counterparty's documentation.

Free, and staying free

The videos, the notebooks and this book are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

The US Dollar Money Market

The Fed sets a range, not a rate. Two overnight benchmarks come out of the market it steers, and US dollar swaps pay the secured one, SOFR.

A range, not a rate

The Federal Reserve's policy committee, the Federal Open Market Committee (FOMC), sets a **target range** for the federal funds rate, 25 basis points wide. Since Thursday 17 September 2026 it has been 3.75% to 4%.

The Fed holds the market inside the range with two rates it administers itself.

- **Interest on reserve balances (IORB)** is what the Fed pays banks on the reserves they keep with it: 3.9%. A bank has little reason to lend overnight for less than it can earn at the Fed.
- **The overnight reverse repo (ON RRP) offering rate**, 3.75%, is the bottom of the range. Here the Fed takes cash overnight against Treasury securities from a wider set of lenders, including money market funds.

Together they put a firm floor under money-market rates and keep the federal funds rate in the range.

The Fed's settings from its Implementation Note of 16 September 2026, and EFFR on the latest day in the data.

SETTING	VALUE
Target range for the federal funds rate	3.75% to 4.00%
Interest on reserve balances (IORB)	3.90%
Overnight reverse repo (ON RRP) offering rate	3.75%
EFFR on the latest day in the data	3.88%

Two overnight rates

Two overnight benchmarks come out of this market every business day, both calculated and published by the Federal Reserve Bank of New York.

	EFFR	SOFR
Market	overnight federal funds: unsecured loans to banks	overnight repo: cash lent against US Treasury securities
Method	volume-weighted median, rounded to the basis point	volume-weighted median, rounded to the basis point
Published	about 9:00am New York time, next business day	about 8:00am New York time, next business day
Calendar	Federal Reserve business days	US government securities market business days

The **effective federal funds rate (EFFR)** measures unsecured overnight borrowing by banks. The **Secured Overnight Financing Rate (SOFR)** measures overnight repo: cash lent against Treasury securities as collateral. The markets are very different in size. On 24 September 2026, EFFR rested on about USD 105 billion of trades and SOFR on about USD 3 trillion, some 28 times as much.

When the Alternative Reference Rates Committee (ARRC) chose a replacement for US dollar LIBOR, it chose SOFR. US dollar overnight index swaps pay SOFR today, and it is the rate the rest of this book builds on.

The two calendars differ too. The data has 12 days with an EFFR but no SOFR, such as Good Friday in 2019: the Fed was open, but the government bond market was closed.

The range since 2018

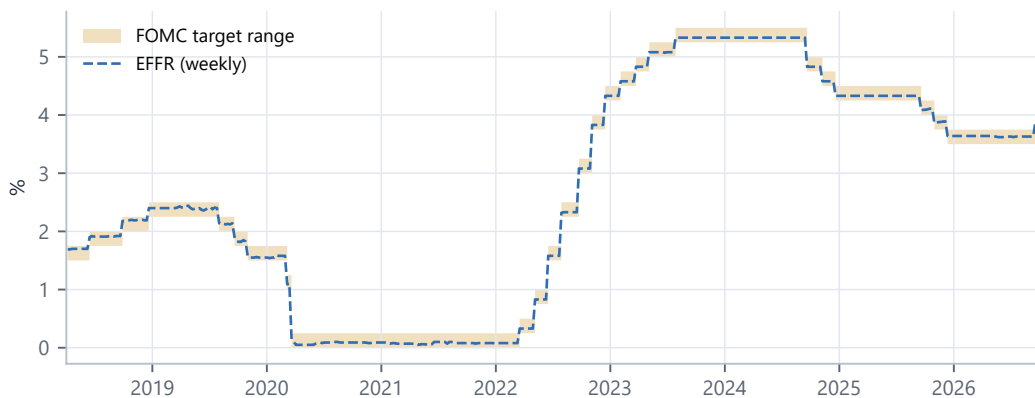


Figure 1.1. The FOMC target range (shaded) and EFFR, weekly, April 2018 to September 2026. Source: Federal Reserve Bank of New York.

The range began the data, in April 2018, at 1.5% to 1.75%, and has changed 26 times since. In March 2020 it was cut to zero to 0.25%. From March 2022 came 11 increases, 525 basis points in all, to 5.25% to 5.5% by July 2023. Then came 6 cuts, from September 2024 to December 2025, down to 3.5% to 3.75%. September 2026 brought the first increase since July 2023.

The range does its job. EFFR printed inside it on 2,130 of the 2,131 days in the data. The one exception was Tuesday 17 September 2019, when EFFR came in at 2.3%, 5 basis points above the top.

The range check: each day's EFFR against that day's range.

```
inside = rates.effr.between(rates.target_low, rates.target_high)
outside = rates[~inside]
o = outside.iloc[0]
```

When repo spikes

SOFR usually sits within about 3 basis points of EFFR, measured as the median absolute difference. But repo can spike.

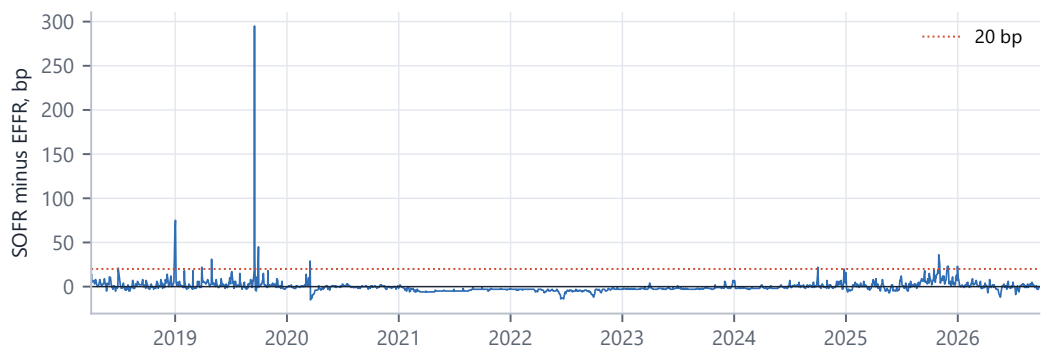


Figure 1.2. SOFR minus EFR, daily, in basis points. The dotted line marks the threshold used below. Source: Federal Reserve Bank of New York.

The largest spike came on that same day, Tuesday 17 September 2019. SOFR fixed at 5.25%, 300 basis points above the top of the range, while EFR was 2.3%. A corporate tax date and large Treasury settlements had drained cash from the repo market.

Smaller spikes cluster at the turn of the month. SOFR was 20 basis points or more above EFR on 17 days, and 12 of them were the first or last business day of a month, although such days are only 10% of the sample. Treasury settlements and reporting dates tend to fall there. The latest run of pressure came between September 2025 and December 2025, when SOFR closed above the top of the range on 17 days.

US and Australian dollar rates side by side

Readers coming from the first course, *AUD Swaps & Curves*, should notice four differences.

	US DOLLAR	AUSTRALIAN DOLLAR
Policy rate	a target range	a single target for the cash rate
Unsecured interbank rate	EFR	AONIA (the cash rate)
Overnight index swaps pay	SOFR, a secured rate	AONIA, an unsecured rate
Money-market day count	actual/360	actual/365 fixed

The last row matters more than it looks. US money markets count interest on a 360-day year. One day's interest on USD 100 million at SOFR's latest 3.88% is USD 10,777.78 on actual/360. Counted on 365, it would be USD 10,630.14. The same quoted rate pays 1.4% more on the US convention. To put a SOFR rate next to an AONIA rate on equal terms, convert it first:

$$r_{365} = r_{360} \times \frac{365}{360}$$

Summary

- The FOMC sets a **target range**; IORB and the ON RRP rate hold the market inside it.
- **EFR** is unsecured; **SOFR** is secured by Treasuries and far larger. US dollar swaps pay **SOFR**.
- EFR has stayed inside the range on every day but one since 2018. SOFR spikes, mostly at month turns.
- US money markets use **actual/360**, so a SOFR rate is not directly comparable with an AONIA rate.

Sources

Companion notebook: `s2e01_usd_money_market.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
2. Alternative Reference Rates Committee, *A User's Guide to SOFR (April 2019)*. https://www.newyorkfed.org/medialibrary/Microsites/arrc/files/2019/Users_Guide_to_SOFR.pdf
3. Bank for International Settlements, *Covered interest parity lost: understanding the cross-currency basis (BIS Quarterly Review, September 2016)*. https://www.bis.org/publ/qtrpdf/r_qt1609e.htm
4. Commodity Futures Trading Commission, *Swaps Made Available To Trade chart (as of 7 July 2023)*. <https://www.cftc.gov/media/8991/swapsmadeavailablechart070723/download>
5. Board of Governors of the Federal Reserve System, *What Happened in Money Markets in September 2019? (FEDS Notes, Anbil, Anderson and Senyuz, 27 February 2020)*. <https://www.federalreserve.gov/econres/notes/feds-notes/what-happened-in-money-markets-in-september-2019-20200227.html>
6. Board of Governors of the Federal Reserve System, *Implementation Note issued September 16, 2026*. <https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a1.htm>
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9. Federal Reserve Bank of New York, *Markets data API: SOFR and EFFR (with FOMC target range and volumes), 2 April 2018 to 24 September 2026*. <https://markets.newyorkfed.org/api/rates/>
10. Federal Reserve Bank of New York, *Additional Information about Reference Rates Administered by the New York Fed*. <https://www.newyorkfed.org/markets/reference-rates/additional-information-about-reference-rates>
11. Federal Reserve Bank of New York, *Secured Overnight Financing Rate Data*. <https://www.newyorkfed.org/markets/reference-rates/sofr>
12. Federal Reserve Bank of New York, *Terms of Use (Reference Rates use restrictions)*. <https://www.newyorkfed.org/privacy/termsfuse>
13. Reserve Bank of Australia, *Cash Rate Target Overview*. <https://www.rba.gov.au/cash-rate-target-overview.html>

How SOFR Compounds

One three-month SOFR swap, settled from the rates the New York Fed actually published, and checked three ways: by hand, in QuantLib, and against the New York Fed's own SOFR Index.

The trade

We receive fixed on a three-month SOFR overnight index swap on USD 100 million, at a fixed rate of 3.6%. The fixed rate is made up for teaching; every SOFR fixing in this chapter is real.

The conventions for US dollar SOFR swaps are:

- **Start:** two New York business days after the trade (T+2).
- **Day count:** actual/360 on both legs.
- **Payments:** both legs pay annually, so a swap of a year or less pays once, at the end.
- **Payment lag:** two business days after the end date.
- **Fixings:** one for each business day of the US government securities market.

The trade was done on Monday 15 June 2026. It starts on Wednesday 17 June 2026 and ends on Thursday 17 September 2026: 92 days. The single payment is on Monday 21 September 2026.

One fixing per bond-market day

The floating leg uses SOFR for every business day of the government securities market from the start date up to, but not including, the end date: 63 fixings here. Each rate r_i accrues for n_i calendar days, until the next such business day. A Friday's rate usually counts three days, and a rate before a holiday counts more.

Selected fixings: the first day, the days that cover long weekends, the highest fixing and the last day. Source: Federal Reserve Bank of New York.

DATE	SOFR (%)	DAYS	$1 + R \cdot N / 360$	COMPOUNDED SO FAR
Wed 17 Jun	3.63	1	1.00010083	1.00010083
Thu 18 Jun	3.62	4	1.00040222	1.00050310
Tue 30 Jun	3.68	1	1.00010222	1.00141083
Thu 2 Jul	3.64	4	1.00040444	1.00191770
Fri 4 Sep	3.65	4	1.00040556	1.00840396
Wed 16 Sep	3.62	1	1.00010056	1.00931858

The fixing calendar is not the Federal Reserve's. On Friday 3 July 2026 the Fed was open but the government securities market was closed, so there is no SOFR for that day, and the Thursday rate covered four days. QuantLib's SOFR index carries the right calendar: the notebook checks that it picks out exactly the

days the New York Fed published.

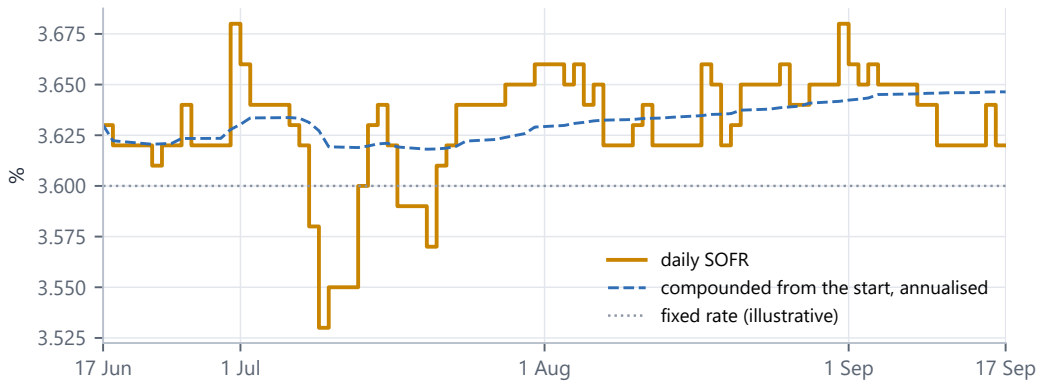


Figure 2.1. Daily SOFR over the life of the swap, SOFR compounded from the start date (annualised), and the fixed rate. Source for SOFR: Federal Reserve Bank of New York.

The highest fixing, 3.68%, fell on 30 June, the end of the quarter. Compounded SOFR finished above the fixed rate.

The formula

The floating amount compounds the daily rates; the fixed amount is simple interest:

$$\text{Float} = N \left[\prod_i \left(1 + r_i \frac{n_i}{360} \right) - 1 \right], \quad \text{Fixed} = N \cdot K \cdot \frac{D}{360}$$

where N is the notional, K the fixed rate and D the number of days from start to end. The two amounts are netted and paid on the payment date.

Compounding by hand.

```
growth = float(fx.factor.prod())
float_hand = notional * (growth - 1)
fixed_amt = notional * K * days / 360
compounded_pct = (growth - 1) * 360 / days * 100
average_pct = float((fx.rate_pct * fx.days).sum() / days)
```

The product of the daily factors is 1.0093185841, so the floating amount is USD 931,858.41. Annualised on actual/360, SOFR compounded to 3.6464%.

What compounding adds

Compounding makes a small difference. The average of the fixings, each weighted by its days, is 3.6299%. Compounded, it is 3.6464%: about 1.7 basis points more, because each day's interest earns interest for the rest of the period. On this swap that is worth USD 4,220.

Three ways, one answer

QuantLib. Load every published fixing into QuantLib's `Sofr` index, build the swap from an explicit schedule with a two-day payment lag, and set the evaluation date to the end date. Every rate is then a past fixing and nothing is forecast.

The same swap in QuantLib, priced from the published fixings.

```
for r in data.dropna(subset=["sofr"]).itertuples():
    sofr.addFixing(ql.Date(r.date.day, r.date.month, r.date.year), r.sofr / 100)

ql.Settings.instance().evaluationDate = end
schedule = ql.Schedule(start, end, ql.Period(ql.Once), pay_cal, ql.ModifiedFollowing, ql.ModifiedFollowing,
                        ql.DateGeneration.Backward, False)
swap = ql.OvernightIndexedSwap(ql.Swap.Receiver, notional, schedule, K, dc, sofr, 0.0,
                               2, ql.Following, pay_cal) # payment lag: two business days
fixed_cf, float_cf = swap.fixedLeg()[0], swap.overnightLeg()[0]
```

The SOFR Index. The New York Fed publishes a SOFR Index that compounds SOFR on every business day since it began at 1.00000000 in April 2018. It is published on its value date and includes that day's SOFR, so the index on the end date divided by the index on the start date is the growth of the floating leg:

$$\text{Float} = N \left(\frac{\text{Index}_{\text{end}}}{\text{Index}_{\text{start}}} - 1 \right) = N \left(\frac{1.25909479}{1.24747013} - 1 \right)$$

	FLOATING AMOUNT	DIFFERENCE FROM THE HAND CALCULATION
By hand	USD 931,858.41	
QuantLib	USD 931,858.41	USD 0.00
SOFR Index	USD 931,858.79	USD 0.37

QuantLib matches the hand calculation to the cent. The index lands within a dollar: it is published to eight decimal places, and on USD 100 million the last decimal is worth a dollar.

Settlement

We receive USD 920,000.00 fixed and owe USD 931,858.41 floating. Compounded SOFR came in above our fixed rate, so we pay the difference, USD 11,858.41, on Monday 21 September 2026. The fixed rate that would have broken even is the compounded rate itself, 3.6464%.

Notice the timing. Each day's SOFR is published the next business day at about 8:00am New York time. The last fixing is for Wednesday 16 September 2026, published on Thursday 17 September 2026, the end date itself. So the floating amount is only known on the last day of the swap, and the two-day payment lag gives both sides time to settle, while interest still accrues over the whole period.

Summary

- A SOFR swap compounds **one fixing per government securities business day**, each weighted by the calendar days it covers, on **actual/360**.
- Compounding adds a little to the day-weighted average.
- The hand calculation, QuantLib and the New York Fed's **SOFR Index** all agree.

- The floating amount is known on the end date and paid **two business days later**.

Sources

Companion notebook: `s2e02_sofr_compounding.ipynb`. Every number in this chapter is computed by it.

1. Alternative Reference Rates Committee, *A User's Guide to SOFR (April 2019)*. https://www.newyorkfed.org/medialibrary/Microsites/arrc/files/2019/Users_Guide_to_SOFR.pdf
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4. Federal Reserve Bank of New York, *Markets data API: SOFR, EFFR, and SOFR Averages and Index*. <https://markets.newyorkfed.org/api/rates/>
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The SOFR Curve

From swap and futures quotes to a whole SOFR curve in one bootstrap, with every quote checked by hand.

SOFR swap conventions

Chapter 2 settled one SOFR swap from past fixings. This chapter goes the other way: from today's market quotes to discount factors for every date out to thirty years. The first ingredient is the swaps. Their conventions differ from Australian dollar swaps in three places.

SOFR swaps against the AONIA swaps of *AUD Swaps & Curves*.

	USD SOFR OIS	AUD AONIA OIS
Start	T+2	T+1
Day count	ACT/360	ACT/365F
Payments	Annual	Annual, front stub
Payment lag	2 days	2 days
Fixing calendar	US govt securities	Sydney

A SOFR swap traded today, Friday 25 September 2026, starts on Tuesday 29 September 2026. Longer swaps pay once a year, with no front stub, each payment two business days after its period ends.

SOFR futures

The short end has a second instrument: **three-month SOFR futures**, listed by CME. Their terms, from CME's listing filing with the CFTC:

- **Reference quarter:** from the third Wednesday of the contract month (an IMM date) up to the third Wednesday three months later. The contract is named after the month its quarter starts.
- **Final settlement:** SOFR compounded over the reference quarter, with the formula of Chapter 2:

$$R = \left[\prod_i \left(1 + \frac{d_i}{360} \frac{r_i}{100} \right) - 1 \right] \times \frac{360}{D} \times 100, \quad \text{price} = 100 - R$$

- **Size:** each basis point of R is worth USD 25 per contract.

Futures settle every day, and swaps don't, which makes the futures rate slightly different from the forward rate of the same quarter. That difference is the **convexity adjustment**. It depends on how volatile rates are, and for contracts a year or so away it is small. In this book we set it to zero, as a simplification.

The four futures in our quotes and their reference quarters. Illustrative prices.

CONTRACT	PRICE	RATE (%)	QUARTER STARTS	QUARTER ENDS	DAYS
SR3 Dec 2026	96.20	3.80	16 Dec 2026	17 Mar 2027	91
SR3 Mar 2027	96.30	3.70	17 Mar 2027	16 Jun 2027	91
SR3 Jun 2027	96.39	3.61	16 Jun 2027	15 Sep 2027	91
SR3 Sep 2027	96.44	3.56	15 Sep 2027	15 Dec 2027	91

The quotes

The curve rests on 16 quotes, all made up for teaching: 12 SOFR swaps and 4 futures. Swaps cover the first three months, the futures step through the next year, and swaps take over from two years to thirty.

The SOFR curve quotes: short swaps and futures on the left, longer swaps on the right. Illustrative, not market data.

INSTRUMENT	KIND	QUOTE	SWAP	RATE
1W	SOFR OIS	3.88%	2Y	3.66%
1M	SOFR OIS	3.87%	3Y	3.63%
2M	SOFR OIS	3.86%	5Y	3.63%
3M	SOFR OIS	3.85%	7Y	3.70%
SR3 Dec 2026	3M SOFR future	96.20	10Y	3.81%
SR3 Mar 2027	3M SOFR future	96.30	15Y	3.94%
SR3 Jun 2027	3M SOFR future	96.39	20Y	3.98%
SR3 Sep 2027	3M SOFR future	96.44	30Y	3.96%

The later contracts have higher prices, so lower rates: these quotes expect SOFR to fall over the next year.

One bootstrap

In QuantLib each quote becomes a rate helper. A SOFR swap is an `OISRateHelper` with the conventions above, a future a `SofrFutureRateHelper` named by its quarter. The curve then solves for one discount factor per quote, shortest first, interpolating log-linearly between them, which means flat forward rates between pillars.

The SOFR curve builder, shared by every later chapter.

```
def sofr_ois_helper(tenor, quote):
    return ql.OISRateHelper(2, ql.Period(tenor), quote, ql.Sofr(), paymentLag=2,
                           paymentFrequency=ql.Annual, paymentCalendar=USD_CAL,
                           convention=ql.ModifiedFollowing, endOfMonth=True)

def build_sofr_curve(quotes):
    """quotes: rows (instrument, Label, value); OIS values in %, futures as prices. Returns curve, quotes,
    helpers."""
    q, helpers = {}, {}
    for r in quotes.itertuples():
        if r.instrument == "future": # Label Like "SR3 Dec 2026"
            q[r.label] = ql.SimpleQuote(r.value)
            _, mon, year = r.label.split()
            helpers[r.label] = ql.SofrFutureRateHelper(ql.QuoteHandle(q[r.label]), MONTHS[mon], int(year),
                                                         ql.Quarterly)
        else:
            q[r.label] = ql.SimpleQuote(r.value / 100)
            helpers[r.label] = sofr_ois_helper(r.label, ql.QuoteHandle(q[r.label]))
    curve = ql.PiecewiseLogLinearDiscount(today, list(helpers.values()), USD_DC)
    curve.enableExtrapolation()
    curve.nodes() # bootstrap now: the helpers only see the curve once it has been built
    return curve, q, helpers
```

Checking it by hand

A future. Compound the curve's forward SOFR day by day over the December quarter with CME's formula. Each day's factor is a ratio of discount factors, $1 + r_i d_i / 360 = P(d_i) / P(d_{i+1})$, so the whole product collapses to the discount factor at the start of the quarter over the one at the end:

$$R = \left(\frac{P(\text{start})}{P(\text{end})} - 1 \right) \frac{360}{D}$$

	RATE (%)
From the price, 100 – 96.20	3.800000
CME's formula, daily forwards	3.800000
Ratio of two discount factors	3.800000

A one-month swap. A swap with a single period has both legs paying on the same day, so the payment date drops out. The fixed rate K must match compounded SOFR over the period, which is the ratio of discount factors:

$$\frac{P(e)}{P(s)} = \frac{1}{1 + K \tau}, \quad \tau = \frac{\text{days}}{360}$$

From 29 Sep 2026 to 29 Oct 2026, 30 days at 3.87%, the hand calculation gives 0.996785367191 and the curve 0.996785367191.

A five-year swap. Longer swaps pay annually, each period two days after it ends. With that lag the floating leg no longer collapses to one ratio, so value each period on its own: forward SOFR compounded over the period, $P(s_i) / P(e_i) - 1$, paid at p_i .

$$K = \frac{\sum_i (P(s_i) / P(e_i) - 1) P(p_i)}{\sum_i \tau_i P(p_i)}$$

The five-year swap, period by period.

START	END	PAID	DAYS/360	FORWARD SOFR (%)	P(PAID)
29 Sep 2026	29 Sep 2027	1 Oct 2027	1.0139	3.7838	0.962450
29 Sep 2027	29 Sep 2028	3 Oct 2028	1.0167	3.5321	0.928913
29 Sep 2028	28 Sep 2029	2 Oct 2029	1.0111	3.5665	0.896576
28 Sep 2029	30 Sep 2030	2 Oct 2030	1.0194	3.6299	0.864748
30 Sep 2030	29 Sep 2031	1 Oct 2031	1.0111	3.6301	0.834132

The par rate comes out at 3.6300% by hand and 3.6300% in QuantLib, against a quote of 3.63%. In fact every one of the 16 quotes reprices to within one billionth of a basis point.

The curve

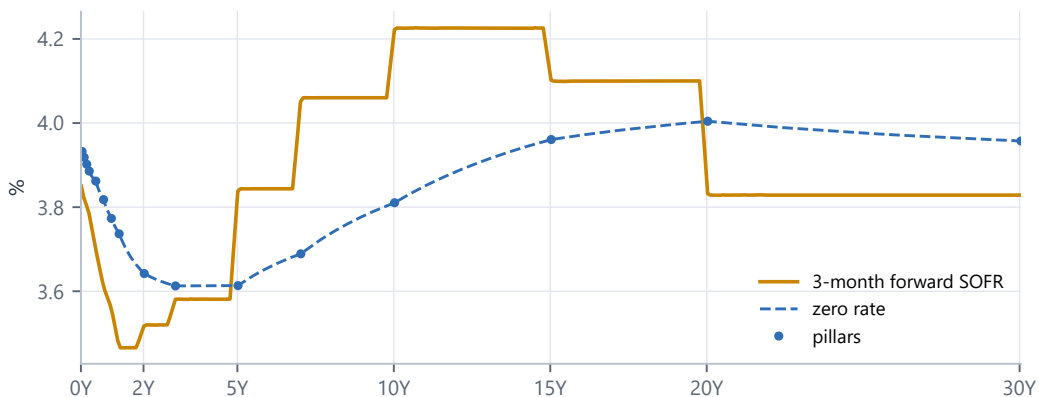


Figure 3.1. Three-month forward SOFR and the zero rate (continuous, actual/365 fixed), with the pillar dates. Illustrative quotes.

Three-month forward SOFR starts at 3.85%, falls to about 3.47% a little over a year out, then climbs to around 4.22% ten years from now. The zero rate averages the forwards, so it moves more gently. The steps come from the flat forwards between pillars: this is what these made-up prices say about future SOFR, and nothing more.

One number for later

The denominator of the par rate formula is the swap's **annuity**, 4.551712 here. On USD 100 million of five-year swaps, one basis point on the fixed rate is worth USD 45,517. Chapter 10 comes back to numbers like this one.

Summary

- SOFR swaps start **T+2**, count days on **actual/360** and pay **annually**, two business days after each period.
- A three-month SOFR future settles on **100 minus compounded SOFR** over its IMM-to-IMM quarter; here its convexity adjustment is set to zero.
- Swaps and futures go into **one bootstrap**, one discount factor per quote.

- A future's rate, and a one-period swap, are each just a **ratio of two discount factors**; longer swaps are valued period by period.

Sources

Companion notebook: `s2e03_sofr_curve.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
2. Commodity Futures Trading Commission, *Swaps Made Available To Trade chart (as of 7 July 2023)*. <https://www.cftc.gov/media/8991/swapsmadeavailablechart070723/download>
3. Chicago Mercantile Exchange, filed with the CFTC, *CME Submission No. 18-069: Initial Listing of the Three-Month SOFR Futures and One-Month SOFR Futures Contracts (19 April 2018)*. <https://www.cftc.gov/sites/default/files/filings/ptc/18/04/ptc041918cmecm002.pdf>
4. Board of Governors of the Federal Reserve System, *Inferring Term Rates from SOFR Futures Prices (FEDS 2019-014, Heitfield and Park)*. <https://www.federalreserve.gov/econres/feds/files/2019014pap.pdf>
5. Federal Reserve Bank of New York, *Terms of Use (Reference Rates use restrictions)*. <https://www.newyorkfed.org/privacy/termsfuse>

AUD/USD Spot

How the exchange rate is quoted, eight years of it from the Federal Reserve Board, how big the market is, and what "spot" really means for settlement.

Reading a quote

AUD/USD is the number of US dollars one Australian dollar buys. The first currency in the pair is the one being priced, the **base** currency; the second is the currency the price is in. A higher number means a stronger Australian dollar. The market quotes AUD/USD to four decimal places, and the fourth decimal, 0.0001, is called a **point** (or a pip).

On Friday 18 September 2026, AUD/USD was 0.7111.

One quote, read three ways. Source: Federal Reserve Board, H.10 release.

	VALUE	MEANING
AUD/USD	0.7111	US dollars for one Australian dollar
USD/AUD	1.4063	Australian dollars for one US dollar
USD 100 million	AUD 140,627,197	divide by AUD/USD

Turning an amount from one currency into the other is where most slips happen. To turn US dollars into Australian dollars, **divide** by AUD/USD; to turn Australian dollars into US dollars, **multiply**. So USD 100 million buys about AUD 140.63m.

Eight years of AUD/USD

The Federal Reserve Board publishes the noon buying rate in New York for the Australian dollar every business day, in its H.10 release.



Figure 4.1. AUD/USD, US dollars per Australian dollar, weekly (Fridays). Source: Board of Governors of the Federal Reserve System, H.10 release.

The data opens on Tuesday 2 January 2018 at 0.7824 and peaks at 0.8105 on Friday 26 January 2018. The low came during the COVID-19 market turmoil, on Monday 23 March 2020: 0.5755. In April 2025 the rate fell to 0.5980 on Monday 7 April 2025, just after its largest one-day move in the data: down 4.9% on Friday 4 April 2025. The latest reading is 0.7111.

A big market

Every three years the Bank for International Settlements surveys foreign exchange trading worldwide; the RBA reported the April 2025 results for Australia.

MEASURE	VALUE
Australian dollar, rank among currencies	7th, around 6% of turnover
AUD/USD, rank among currency pairs	6th, around 5% of turnover
Deals with the US dollar on one side	around 90%
FX swaps, share of all turnover	42%

The US dollar is on one side of most deals, which is why so much of this book runs through it. And the most traded instrument is not spot but the **FX swap**, the subject of Chapter 5.

(Currency shares add up to 200% across all currencies, because every deal has two.)

Spot means two days

A **spot** deal is not settled today. It settles within two business days, on the **spot date**. For AUD/USD, the market counts two days that are good business days in **both** Sydney and New York. A holiday in either city pushes the spot date out.

Spot dates around Sydney and New York holidays in late 2026. Holidays checked against the RBA's list of public and bank holidays.

TRADE	SPOT	CALENDAR DAYS	WHY
Mon 28 Sep	Wed 30 Sep	2	no holidays
Fri 2 Oct	Wed 7 Oct	5	Monday 5 October is Labour Day in Sydney
Thu 8 Oct	Tue 13 Oct	5	Monday 12 October is Columbus Day in New York
Mon 9 Nov	Thu 12 Nov	3	Wednesday 11 November is Veterans Day in New York

In QuantLib the spot date is one call: advance two business days on a **joint calendar**, closed whenever either city is.

Spot dates on a joint Sydney and New York calendar.

```
syd = ql.Australia(ql.Australia.Settlement)
ny = ql.UnitedStates(ql.UnitedStates.Settlement)
joint = ql.JointCalendar(syd, ny)
RBA_HOLIDAYS_2026 = {"2026-10-05": "Labour Day, NSW", "2026-10-12": "Columbus Day, New York",
                    "2026-11-11": "Veterans Day, New York"}
for d, name in RBA_HOLIDAYS_2026.items():
    qd = ql.DateParser.parseISO(d)
    assert not joint.isBusinessDay(qd), f"QuantLib misses {name}"

def spot(trade):
    return joint.advance(trade, 2, ql.Days)
```

The same function reproduces AFMA's own worked example: a trade on Friday 26 April 2024 settles spot on Tuesday 30 April 2024.

One case left out. When the day after the trade is a holiday in New York only, the FX market has its own rule for whether that day counts. QuantLib's `FxSwapRateHelper` has a `tradingCalendar` option for it. None of the examples in this book falls on that case.

Why the spot date matters

Everything that follows is measured from spot. FX forwards and swaps in Chapter 5 are quoted for periods starting on the spot date, the cross-currency swaps of Chapter 6 start there, and the curves of Chapter 8 are anchored there. Getting spot wrong by a day shifts every date that hangs from it.

Summary

- AUD/USD is **US dollars per Australian dollar**; a point is 0.0001.
- To convert US dollars to Australian dollars, **divide** by AUD/USD.
- The Australian dollar is one of the world's most traded currencies, and **FX swaps** are the most traded FX instrument.
- **Spot** is two good business days in **both** Sydney and New York: a joint calendar in QuantLib.

Sources

Companion notebook: `s2e04_audusd_spot.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
2. Bank for International Settlements, *Triennial Central Bank Survey 2025: reporting guidelines for turnover*. <https://www.bis.org/publications/triennial-survey/2025survey-guidelinesturnover.pdf>
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7. Reserve Bank of Australia, *Developments in Foreign Exchange and Over-the-counter Derivatives Markets (Bulletin, February 2026)*. <https://www.rba.gov.au/publications/bulletin/2026/feb/developments-in-foreign-exchange-and-over-the-counter-derivatives-markets.html>
8. Reserve Bank of Australia, *Public & Bank Holidays 2026*. <https://www.rba.gov.au/schedules-events/bank-holidays/bank-holidays-2026.html>

FX Forwards and FX Swaps

Everything after spot. Forward points, covered interest parity, and the Australian dollar interest rate the FX market implies, which turns out not to be AONIA.

Forwards and FX swaps

An **outright forward** is an exchange rate agreed today for settlement on a date after spot. An **FX swap** pairs two exchanges: the currencies change hands on one date (the **near leg**, usually spot) and change back on a later date (the **far leg**), both at rates agreed today. FX swaps are the most traded instrument in the whole foreign exchange market, 42% of turnover.

Here is one. An Australian bank needs US dollars for three months. On the spot date it hands over AUD 100 million and receives US dollars at spot, 0.7100. Three months later it gets its Australian dollars back and returns the US dollars at the three-month forward rate, 0.709571.

A three-month FX swap on AUD 100 million, seen from the bank. Illustrative rates.

LEG	DATE	AUSTRALIAN DOLLARS	US DOLLARS
Near (spot)	29 Sep 2026	pay AUD 100,000,000	receive USD 71,000,000.00
Far (3 months)	29 Dec 2026	receive AUD 100,000,000	pay USD 70,957,100.00

The bank has, in effect, borrowed US dollars and lent Australian dollars for three months, secured by each other, with no currency risk: both exchange rates were fixed on day one. The difference between the two US dollar amounts plays the part of interest.

Forward points

The market quotes forwards as **points**: the forward rate minus spot, in units of the fourth decimal place.

$$F = S + \frac{\text{points}}{10,000}$$

Illustrative AUD/USD forward points and outright forwards, spot 0.7100. Not market data.

TENOR	VALUE DATE	POINTS	OUTRIGHT
Spot	29 Sep 2026	0.00	0.710000
1W	6 Oct 2026	-0.23	0.709977
1M	29 Oct 2026	-1.10	0.709890
2M	30 Nov 2026	-2.66	0.709734
3M	29 Dec 2026	-4.29	0.709571
6M	30 Mar 2027	-10.68	0.708932
9M	29 Jun 2027	-19.65	0.708035
1Y	29 Sep 2027	-30.32	0.706968

With spot at 0.7100, the one-year points are -30.32, so the one-year outright is 0.7070. Every forward is below spot, and the further out, the lower.

Two more quotes cover the days *before* spot. The **overnight** swap runs from today to tomorrow, the **tom-next** swap from tomorrow to spot. Their points, -0.098 and -0.033, walk the rate back from spot to today: the rate for settlement today is 0.7100131.

Covered interest parity

Why below spot? Borrowing US dollars directly should cost the same as borrowing Australian dollars and swapping them into US dollars; otherwise there is a riskless profit. That is **covered interest parity**. For AUD/USD, from spot s to T :

$$F = S \times \frac{1 + r_{\text{USD}} \tau_{360}}{1 + r_{\text{AUD}} \tau_{365}} = S \times \frac{P_{\text{AUD}}(s, T)}{P_{\text{USD}}(s, T)}$$

Each rate uses its own day count, and the second form writes each growth factor as a ratio of discount factors. The currency with the higher interest rate trades at a discount in the forward market, just enough to cancel its extra interest.

Try it for three months (91 days) with the SOFR curve of Chapter 3 and the AONIA curve from *AUD Swaps & Curves*:

	VALUE
SOFR, simple, actual/360	3.8500%
AONIA, simple, actual/365	3.8900%
Forward from parity	0.710024 (0.24 points)
Forward in the market	0.709571 (-4.29 points)

Once the day counts are allowed for, SOFR and AONIA are almost the same, so parity puts the forward right on spot. The market's forward is 4.53 points lower.

The rate the FX market implies

Turn parity around. Given the market's forward and SOFR, which Australian dollar rate makes it hold?

$$r_{\text{AUD}}^{\text{implied}} = \left[\frac{S}{F} (1 + r_{\text{USD}} \tau_{360}) - 1 \right] \frac{365}{\text{days}}$$

At three months it is 4.1483%, against AONIA at 3.8900%: 25.8 basis points higher.

The Australian dollar rate implied by FX swaps and SOFR, against AONIA, simple rates from spot. Illustrative quotes.

TENOR	DAYS	SOFR (%)	AONIA (%)	AUD IMPLIED (%)	GAP (BP)
1W	7	3.8800	3.8553	4.1029	24.8
1M	30	3.8700	3.8600	4.1129	25.3
2M	62	3.8600	3.8798	4.1357	25.6
3M	91	3.8500	3.8900	4.1483	25.8
6M	182	3.8353	3.9300	4.1966	26.7
9M	273	3.8097	3.9704	4.2444	27.4
1Y	365	3.7838	4.0003	4.2816	28.1

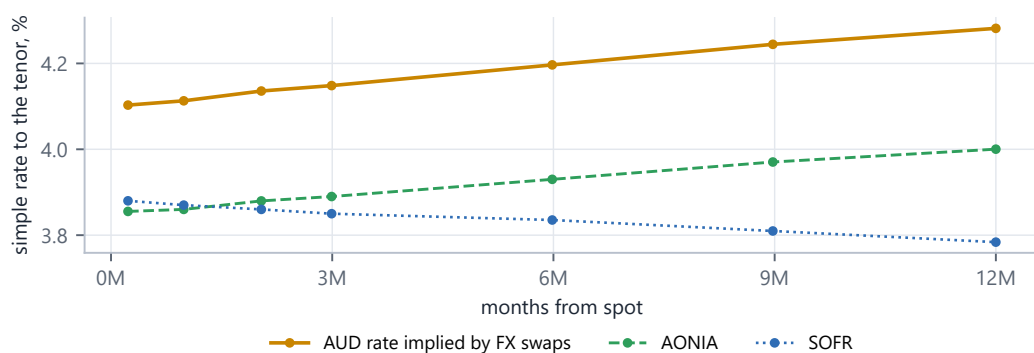


Figure 5.1. SOFR, AONIA and the Australian dollar rate implied by FX swaps, from spot to each tenor. Illustrative quotes.

The gap is there at every tenor, 24.8 to 28.1 basis points in our quotes, widening a little with maturity.

The cross-currency basis

In textbooks that gap shouldn't exist. Since the 2007–08 financial crisis it has, in most currency pairs, and it has a name: the **cross-currency basis**. It is the extra cost, above the interest rates themselves, of borrowing one currency through an FX swap in the other.

For the Australian dollar the basis is usually positive, and the reason is funding. Australian banks raise much of their funding offshore, in US dollars and other currencies, and swap it back into Australian dollars. That steady demand for Australian dollars in the swap market pushes up the rate implied for them. Our numbers are made up, but the sign is the real one.

In QuantLib

QuantLib's `FxSwapRateHelper` turns each quote into a condition on a curve. It holds the SOFR curve fixed and solves for the Australian dollar discount factors that reproduce the points. The overnight and tom-next helpers have a one-day tenor, with 0 and 1 fixing days, and their near rates are built back from spot.

An Australian dollar curve from FX swaps, given SOFR.

```
near_tn = ql.SimpleQuote(S - points["TN"] / 1e4)
near_on = ql.SimpleQuote(S - (points["TN"] + points["ON"]) / 1e4)
spot_q = ql.SimpleQuote(S)
helpers = [
    ql.FxSwapRateHelper(ql.QuoteHandle(ql.SimpleQuote(points["ON"] / 1e4)), ql.QuoteHandle(near_on),
        ql.Period(1, ql.Days), 0, joint, ql.Following, False, False, sofr),
    ql.FxSwapRateHelper(ql.QuoteHandle(ql.SimpleQuote(points["TN"] / 1e4)), ql.QuoteHandle(near_tn),
        ql.Period(1, ql.Days), 1, joint, ql.Following, False, False, sofr),
] + [ql.FxSwapRateHelper(ql.QuoteHandle(ql.SimpleQuote(points[t] / 1e4)), ql.QuoteHandle(spot_q),
    ql.Period(t), 2, joint, ql.ModifiedFollowing, True, False, sofr)
    for t in ["1W", "1M", "2M", "3M", "6M", "9M", "1Y"]]
aud_fx = ql.PiecewiseLogLinearDiscount(today, helpers, dc)
aud_fx.nodes()
```

The flag `isFxBaseCurrencyCollateralCurrency=False` says the Australian dollar, the base currency of AUD/USD, is not the collateral currency: the US dollar curve is the one held fixed. Chapter 7 explains why that matters. The curve reprices every quote to within one hundred-billionth of a point, and its three-month rate is the same 4.1483% found by hand.

Summary

- An **FX swap** exchanges currencies on one date and back on another; forwards are quoted as **points** on spot.
- **Covered interest parity** ties the forward to the two currencies' discount factors: $F = S \times P_{\text{AUD}}/P_{\text{USD}}$.
- Turned around, FX swaps imply an Australian dollar rate, here some 26 basis points **above AONIA**.
- That gap is the **cross-currency basis**, usually positive for the Australian dollar.

Sources

Companion notebook: `s2e05_fx_forwards_swaps.ipynb`. Every number in this chapter is computed by it.

1. Bank for International Settlements, *Covered interest parity lost: understanding the cross-currency basis* (BIS Quarterly Review, September 2016). https://www.bis.org/publ/qtrpdf/r_qt1609e.htm
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5. Reserve Bank of Australia, *The Financial Crisis through the Lens of Foreign Exchange Swap Markets* (Bulletin, June 2010). <https://www.rba.gov.au/publications/bulletin/2010/jun/7.html>
6. Reserve Bank of Australia, *OTC Derivatives Reforms and the Australian Cross-currency Swap Market* (Bulletin, June 2013). <https://www.rba.gov.au/publications/bulletin/2013/jun/7.html>
7. Reserve Bank of Australia, *Foreign Currency Exposure and Hedging in Australia* (Bulletin, March 2023). <https://www.rba.gov.au/publications/bulletin/2023/mar/foreign-currency-exposure-and-hedging-in-australia.html>
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Cross-Currency Basis Swaps

The swap that trades the cross-currency basis for years at a time. BBSW plus a spread against SOFR, with principals exchanged and a US dollar notional that resets every quarter.

The contract

Chapter 5 ended on a gap: FX swaps imply an Australian dollar rate above AONIA. FX swaps rarely go beyond a year or so. For longer terms the basis trades in a swap of its own, built for hedging long-term foreign borrowing: the **cross-currency basis swap**.

In Australia the standard contract is a BBSW against SOFR basis swap, and AFMA sets out its terms. One side receives three-month BBSW **plus a spread**, the basis, on a constant Australian dollar notional. The other receives SOFR, compounded daily with no spread, on a US dollar notional.

The main terms of a BBSW/SOFR cross-currency basis swap, from AFMA's term sheet.

	AUD LEG	USD LEG
Floating rate	3M BBSW + spread	SOFR, compounded, no spread
Day count	ACT/365F	ACT/360
Payments	Quarterly, 2 days after	Quarterly, 2 days after
Notional	Constant	Reset each quarter at FX
Principal	Exchanged at start and end	Exchanged at start and end
Start	T+2, Sydney and New York	T+2, Sydney and New York

Two things set it apart from the single-currency swaps of *AUD Swaps & Curves*. The **principals are exchanged**, at the start and again at the end, because the two sides are really lending each other money in two currencies. And the swap is **not centrally cleared**: it is traded bilaterally, usually under a collateral agreement, which becomes important in Chapter 7.

A two-year swap, quarter by quarter

Take a two-year swap on AUD 100 million at a basis of 14.75 basis points, seen from the side that receives the Australian dollar leg. At the start it pays AUD 100 million and receives USD 71.00m, the Australian notional at today's rate, 0.7100. Then, each quarter:

- The Australian dollar leg earns BBSW plus the basis on AUD 100 million.
- The US dollar leg costs SOFR, compounded over the quarter, on the US dollar notional.
- The US dollar notional is **reset** to the Australian notional times the FX rate of the day, and the difference changes hands.

We don't know future FX rates, so the forward rate for each reset date stands in for it, from the curve built in Chapter 8. Forward BBSW and SOFR come from their curves.

The two-year swap from the side receiving the AUD leg: forward rates, forward FX at each reset, and coupons. Illustrative quotes.

QTR	STARTS	BBSW + BASIS (%)	AUD COUPON	FX AT RESET	USD NOTIONAL	SOFR (%)	USD COUPON
1	29 Sep 2026	4.1686	1,062,125	0.710000	USD 71.00m	3.8496	706,086
2	31 Dec 2026	4.2360	1,044,504	0.709558	USD 70.96m	3.7820	670,886
3	31 Mar 2027	4.2363	1,056,166	0.708922	USD 70.89m	3.6862	660,558
4	30 Jun 2027	4.2400	1,068,714	0.708024	USD 70.80m	3.6020	651,748
5	30 Sep 2027	4.3976	1,096,383	0.706953	USD 70.70m	3.5444	633,398
6	30 Dec 2027	4.3978	1,108,493	0.705542	USD 70.55m	3.4658	624,895
7	31 Mar 2028	4.3976	1,096,383	0.703977	USD 70.40m	3.4656	616,702
8	30 Jun 2028	4.3986	1,096,635	0.702433	USD 70.24m	3.4656	615,350

In the first quarter the Australian dollar side earns 4.17% and the US dollar side pays 3.85%. Because forward AUD/USD falls over time, the US dollar notional drifts down, to USD 70.24m in the last quarter. At the first reset alone it falls by USD 44,189.

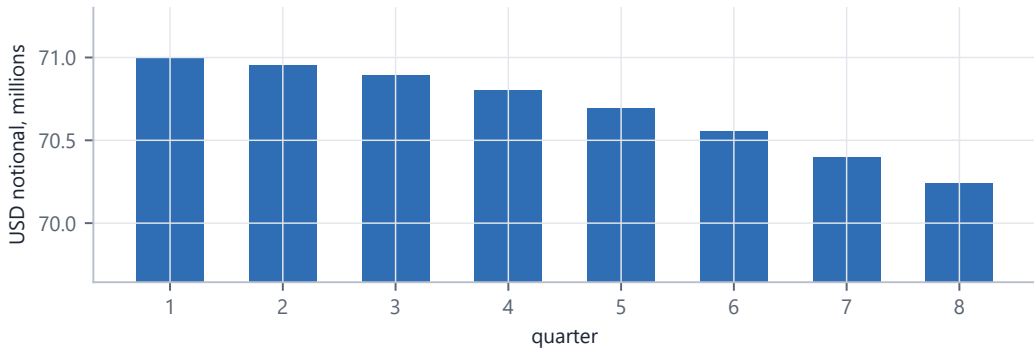


Figure 6.1. The US dollar notional of the two-year swap, quarter by quarter, reset at forward FX rates. Illustrative quotes.

Why reset?

A scenario, not a forecast: suppose AUD/USD rises 10% soon after the start, from 0.7100 to 0.7810.

- **Without resets**, the USD 71 million to be handed back at the end is now worth only AUD 90.91m. One side is AUD 9.09m ahead and the other owes it, for the rest of the swap's life.

- **With resets**, the US dollar notional moves to USD 78 million at the next reset and the difference is paid over then. Only the move since the last reset is ever outstanding.

Resets cut the counterparty exposure that a long cross-currency swap would otherwise build up. The RBA reports that most swaps entered into by Australian banks reset; the ARRC calls resets a key tool for reducing counterparty exposure.

Valuing it

From the side receiving the Australian dollar leg, in Australian dollars today, with X_0 today's rate and X_i the forward FX rate at the start of quarter i :

$$V = \underbrace{N[P_A(t_n) - P_A(t_0)] + \sum_i N(L_i + s) \tau_i P_A(p_i)}_{\text{AUD leg}} + \underbrace{\frac{1}{X_0} \sum_i N X_i [P_U(t_{i-1}) - P_U(t_i) - R_i \tau'_i P_U(p_i)]}_{\text{USD leg, in USD}}$$

where L_i is forward BBSW, R_i compounded SOFR, s the basis, p_i the payment date and P_A , P_U the Australian and US dollar discount curves. Each quarter the US dollar leg looks like a fresh loan: receive the notional at the start, repay it at the end with SOFR interest.

Which Australian dollar curve is P_A ? Not AONIA: it is the curve of Chapter 8, for Australian dollar cash flows under US dollar collateral. The next two chapters explain why, and build it.

The swap as a loop over quarters. The same function values basis swaps in Chapters 8, 9 and 10.

```
def mtm_swap(m, tenor, spread, aud_notional, coupon_lag=0):
    """Value a BBSW/SOFR mark-to-market basis swap to the party receiving the AUD Leg (BBSW + spread).

    Dates as the curve's helpers build them: spot + tenor, rolled backwards quarterly, end-of-month rule.
    Each quarter the USD notional is the AUD notional times the forward FX rate to the period start; notional
    move on roll dates; coupons are paid coupon_lag business days after each period end (term sheet: 2; the
    curve build: 0). Values are in AUD at today's FX rate. Returns the value, the value of one basis point of
    spread, and one row per period.
    """
    sch = list(q1.MakeSchedule(m.spot, m.spot + q1.Period(tenor), q1.Period(q1.Quarterly), calendar=JOINT,
                              convention=q1.ModifiedFollowing, endOfMonth=True, backwards=True))
    pa, pu = m.aud_usd.discount, m.sofr.discount
    fx_today = m.fwd_fx(today)
    index = bbsw(3, m.bbsw3m)
    pv_aud = -aud_notional * pa(sch[0]) + aud_notional * pa(sch[-1])
    pv_usd, bps, rows = 0.0, 0.0, []
    for i, (a, b) in enumerate(zip(sch[:-1], sch[1:])):
        pay = JOINT.advance(b, coupon_lag, q1.Days) if coupon_lag else b
        tau_a, tau_u = dc.yearFraction(a, b), USD_DC.yearFraction(a, b)
        bbsw_rate = (m.bbsw3m.discount(a) / m.bbsw3m.discount(b) - 1) / tau_a
        sofr_rate = (pu(a) / pu(b) - 1) / tau_u
        fx = m.fwd_fx(a)
        usd_n = aud_notional * fx
        aud_cpn = aud_notional * (bbsw_rate + spread) * tau_a
        usd_cpn = usd_n * sofr_rate * tau_u
        pv_aud += aud_cpn * pa(pay)
        bps += aud_notional * tau_a * pa(pay) * 1e-4
        pv_usd += usd_n * (pu(a) - pu(b)) - usd_cpn * pu(pay) # receive the USD notional, repay it with SOFR
        rows.append({"period": i + 1, "start": a.ISO(), "end": b.ISO(), "pay": pay.ISO(),
                    "bbsw_pct": bbsw_rate * 100, "aud_rate_pct": (bbsw_rate + spread) * 100, "aud_coupon":
                    "fx": fx, "usd_notional": usd_n, "sofr_pct": sofr_rate * 100, "usd_coupon": usd_cpn})
    aud_cpn,
    value = pv_aud + pv_usd / fx_today
    return {"value": value, "bps": bps, "pv_aud_leg": pv_aud, "pv_usd_leg_aud": pv_usd / fx_today, "rows": rows}
```

Who pays the basis

Australian banks raise a large share of their funding offshore and swap it back into Australian dollars. In that swap they receive the foreign currency and **pay the Australian dollar leg**, so they pay the basis, which for AUD/USD is usually positive: it is the cost of hedging their foreign debt. On the other side are foreign borrowers who issue Australian dollar bonds (the "Kangaroo" market) and swap the proceeds into their own currency, receiving the basis.

How big the basis gets varies with funding conditions. For US dollar issuance, the RBA reported it widening from around 5–10 basis points before the financial crisis to around 30–40 by early 2010.

Quotes

The market quotes the basis as the spread on the Australian dollar leg, on the assumption that collateral is posted in US dollars and discounted at SOFR.

Illustrative BBSW/SOFR basis quotes, spread on the AUD leg, US dollar collateral. Not market data.

TENOR	BASIS (BP)
2Y	14.75
3Y	14.75
4Y	15.25
5Y	15.50
7Y	15.50
10Y	15.50

Our made-up quotes run from 14.75 to 15.5 basis points. That collateral assumption turns out to matter a great deal, and it is the subject of the next chapter.

Summary

- A **BBSW/SOFR basis swap** exchanges BBSW plus the basis for SOFR, with **principals exchanged** at both ends.
- The **US dollar notional resets** every quarter at the FX rate of the day, so only one quarter's FX move is ever outstanding.
- Australian banks hedging offshore debt **pay the basis**; foreign issuers of Australian dollar bonds receive it.
- Quotes are spreads on the AUD leg, assuming **US dollar collateral**.

Sources

Companion notebook: `s2e06_xccy_basis_swaps.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
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9. Reserve Bank of Australia, *Australia's External Position and the Evolution of the FX Markets (speech, 29 April 2025)*. <https://www.rba.gov.au/speeches/2025/sp-ag-2025-04-29.html>

Collateral and Discounting

The currency of the collateral decides which curve discounts a cash flow. The same Australian dollar payment has two different values, depending on whether it is secured with Australian or US dollars.

Collateral sets the discount rate

Most swaps between banks are **collateralised**. As a trade's value moves, the side that is behind posts cash to the other, usually every day, so the trade stays almost fully secured. Whoever holds the cash pays interest on it at the rate set in the collateral agreement, usually the overnight rate of its currency: AONIA for Australian dollars, SOFR for US dollars.

Follow the money and a striking result appears. Holding a fully collateralised trade is like holding an asset financed entirely by the collateral: every dollar of value is matched by a dollar of collateral that earns, or costs, the collateral rate. So the value of the trade must grow at the collateral rate, which means a collateralised cash flow is **discounted at the collateral rate**. For a cash flow X at time T , collateralised in its own currency:

$$PV = X \cdot P_c(T), \quad P_c(T) = \text{the discount factor from the overnight rate of the collateral}$$

This is the reason overnight index swap curves, AONIA and SOFR, became the discount curves of the swap market. Fujii, Shimada and Takahashi work it out formally.

Collateral in another currency

What if an Australian dollar cash flow is collateralised in **US dollars**? Then the discount rate has to come from the US dollar collateral rate, carried across by the FX forward. Turn the Australian dollars into US dollars at the forward rate for the payment date, discount in US dollars at SOFR, and turn the result back into Australian dollars at today's rate:

$$PV_{AUD} = X \cdot \frac{F(T) P_{SOFR}(T)}{F(\text{today})} \equiv X \cdot P_{AUD|USD}(T)$$

The ratio defines a new discount curve, $P_{AUD|USD}$: the **Australian dollar curve under US dollar collateral**. Compare it with covered interest parity in Chapter 5, $F = S \times P_{AUD}/P_{USD}$. It is the same equation, solved for the Australian dollar discount factor. So this curve is the one the FX market implies, and its rate is the rate we found above AONIA in Chapter 5.

Which curve for which trade

Discounting by collateral currency.

TRADE	COLLATERAL	DISCOUNT CURVE
AUD swaps	AUD	AONIA
USD swaps (cleared)	USD	SOFR
AUD/USD basis swaps	USD	SOFR; AUD under USD collateral

- **Australian dollar swaps** are discounted on AONIA, cleared or not (AFMA).
- **US dollar swaps** cleared at a clearing house such as LCH are discounted on SOFR.
- **AUD/USD basis swaps** are quoted assuming US dollar collateral: the US dollar leg is discounted on SOFR, and the Australian dollar leg on $P_{\text{AUD}|\text{USD}}$, which is built in Chapter 8.

One cash flow, two values

Take AUD 10 million paid on Wednesday 29 September 2027, a year after spot. The one-year FX swap quote gives the forward, 0.706968; spot less the tom-next and overnight points gives today's rate, 0.7100131.

	AUSTRALIAN DOLLAR COLLATERAL	US DOLLAR COLLATERAL
Discount factor	0.961130 (AONIA)	0.958511
Value today	AUD 9,611,302	AUD 9,585,107
Rate, simple, actual/365	4.00%	4.28%

Same cash flow, same counterparty, different collateral: AUD 26,195 less under US dollar collateral, because the rate the FX market implies is above AONIA. The discount factor from the FX quotes matches the full curve of Chapter 8 to within one ten-trillionth.

Out to ten years

Using the whole curve built in the next chapter, the gap grows with maturity.

AUD 10 million paid on each date, valued under Australian and under US dollar collateral. Illustrative quotes.

YEARS	PAID	P, AONIA	P, USD COLLATERAL	DIFFERENCE ON AUD 10M	ZERO RATE GAP (BP)
1	29 Sep 2027	0.961130	0.958511	AUD 26,195	27.0
2	29 Sep 2028	0.922613	0.917332	AUD 52,804	28.5
5	29 Sep 2031	0.811710	0.799279	AUD 124,315	30.8
10	29 Sep 2036	0.640904	0.620794	AUD 201,098	31.8

The zero rate gap widens from about 27 basis points at one year to 32 at ten, and AUD 10 million paid in ten years is worth AUD 201,098 less under US dollar collateral. For a large book, getting the collateral wrong is a real mispricing.

In QuantLib

In QuantLib the cash flows don't change at all; only the discount curve handed to the pricer does.

The same four cash flows under two discount curves.

```
flows = [ql.SimpleCashFlow(X, JOINT.advance(m.spot, ql.Period(y, ql.Years), ql.ModifiedFollowing, True))
         for y in (1, 2, 5, 10)]
pv_aonia = ql.CashFlows.npv(flows, m.aonia, False, today, today)
pv_usd = ql.CashFlows.npv(flows, m.aud_usd, False, today, today)
out["quantlib"] = {"pv_aonia": pv_aonia, "pv_usd_collateral": pv_usd, "difference": pv_aonia - pv_usd,
```

The four cash flows together are worth AUD 33,363,567 under Australian dollar collateral and AUD 32,959,155 under US dollar collateral. The difference, AUD 404,412, is the sum of the differences in the table.

Summary

- A fully collateralised cash flow is **discounted at the collateral rate**.
- Australian dollar swaps use **AONIA**, cleared US dollar swaps use **SOFR**.
- Australian dollar cash flows under **US dollar collateral** use $P_{\text{AUD}|\text{USD}} = F(T) P_{\text{SOFR}}(T) / F(\text{today})$, the curve the FX market implies.
- The gap to AONIA is some 27 to 32 basis points in our quotes, enough to matter on any sizeable book.

Sources

Companion notebook: `s2e07_collateral.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Interest Rate Derivative Conventions (June 2025)*. <https://www.afma.com.au/getattachment/Standards/Market-Conventions/Sections/Content/Interest-Rate-Derivative-Conventions-2025-06.pdf?lang=en-AU>
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Building the Curve

One Australian dollar discount curve for trades collateralised in US dollars, from overnight to ten years: FX swaps at the short end, BBSW/SOFR basis swaps at the long end, and one bootstrap.

Inputs and the one unknown

Everything else is already in place:

- the **SOFR curve** of Chapter 3;
- the **AONIA and 3-month BBSW curves** from *AUD Swaps & Curves*;
- today's **spot rate**, 0.7100.

To those we add 15 quotes, all made up for teaching: 9 FX swaps from overnight to one year, and 6 basis swaps from two years to ten.

The curve's quotes: AUD/USD forward points and BBSW/SOFR basis spreads. Illustrative, not market data.

FX SWAP	POINTS	BASIS SWAP	BASIS (BP)
ON	-0.098	2Y	14.75
TN	-0.033	3Y	14.75
1W	-0.23	4Y	15.25
1M	-1.10	5Y	15.50
2M	-2.66	7Y	15.50
3M	-4.29	10Y	15.50
6M	-10.68		
9M	-19.65		
1Y	-30.32		

The one unknown is $P_{\text{AUD}|\text{USD}}$, the Australian dollar discount curve under US dollar collateral from Chapter 7. Each quote pins down one discount factor on it.

What each quote says

An **FX swap** fixes a forward, and by covered interest parity the forward is spot times the ratio of the two currencies' discount factors. With SOFR known, the only unknown is the Australian discount factor at the swap's date:

$$F(t) = S \times \frac{P_{\text{AUD}}(s, t)}{P_{\text{SOFR}}(s, t)}$$

A **basis swap** says its two legs are worth the same: BBSW plus the basis on a constant Australian notional, discounted on the Australian curve, against SOFR on a notional that resets at the forward FX rate, discounted on SOFR (the formula in Chapter 6).

$$V_{\text{AUD leg}}(P_{\text{AUD}}) = V_{\text{USD leg}}(P_{\text{SOFR}}, F)$$

Given the curve up to the previous pillar, the only unknown is the Australian discount factor at the swap's maturity. So the bootstrap runs from the shortest quote to the longest, one discount factor at a time, exactly as for the SOFR curve.

In QuantLib

The FX swaps use the `FxSwapRateHelper` of Chapter 5. Each basis swap is a `MtMCrossCurrencyBasisSwapRateHelper`, where "MtM" (mark-to-market) means the resetting notional. Three flags set it up for this market:

FLAG	VALUE	MEANING
<code>isFxBaseCurrencyCollateralCurrency</code>	<code>False</code>	the collateral is US dollars, not the base currency (AUD)
<code>isBasisOnFxBaseCurrencyLeg</code>	<code>True</code>	the spread is on the Australian dollar leg
<code>isFxBaseCurrencyLegResettable</code>	<code>False</code>	the US dollar notional resets; the Australian one is constant

The basis swap helpers and the bootstrap, from the shared 'Market' class.

```
for t, q in self.basis.items():
    self.helpers["basis " + t] = q1.MtMCrossCurrencyBasisSwapRateHelper(
        h(q), q1.Period(t), 2, JOINT, q1.ModifiedFollowing, True,
        bbsw(3, self.bbsw3m), q1.Sofr(self.sofr), self.sofr,
        False, # isFxBaseCurrencyCollateralCurrency: AUD is the base currency; collateral is USD
        True,  # isBasisOnFxBaseCurrencyLeg: the spread is on the AUD (BBSW) leg
        False, # isFxBaseCurrencyLegResettable: the USD notional resets, the AUD one is constant
        q1.Quarterly, 0, q1.Quarterly)
self.curve = q1.PiecewiseLogLinearDiscount(today, list(self.helpers.values()), dc)
self.curve.enableExtrapolation()
self.curve.nodes()
self.aud_usd = q1.YieldTermStructureHandle(self.curve)
```

The helper is handed the 3-month BBSW index, projected on its curve from *AUD Swaps & Curves*, and the SOFR index and discount curve from Chapter 3. One bootstrap then solves for all 15 discount factors.

Checking it without the helper

A curve is only as good as its checks. Three of them:

Checks on the finished curve.

CHECK	RESULT
FX swaps reprice (largest error, pips)	2.1e-12
Basis swaps reprice (largest error, bp)	4.9e-10
5Y basis swap at the quote, by hand (AUD)	0.00
1Y forward: curve vs quote	0.706968 vs 0.706968

The most telling is the second-last. A five-year basis swap on AUD 100 million at the quoted basis, valued leg by leg, quarter by quarter, with our own loop from Chapter 6 rather than QuantLib's helper, is worth zero to the cent. And the one-year forward read off the curve lands exactly on spot plus the one-year points.

One modelling choice

AFMA's term sheet pays each coupon **two business days after** its period, but exchanges the notionals **on the roll dates**. QuantLib's helper puts coupons and notional exchanges on the same dates, so it can't do both. There are two ways to set it:

- **Payment lag 0:** notionals on the roll dates, as in the term sheet; coupons two days early.
- **Payment lag 2:** coupons on time; notionals two days late.

To see which matters less, build the curve both ways, then price every quoted swap exactly as the term sheet says and compare its fair spread with the quote.

Fair spreads under the exact term-sheet dates, on curves built with each payment lag.

HELPER	LAG	TENOR	QUOTE (BP)	TERM-SHEET FAIR SPREAD (BP)
	0	2Y	14.75	14.8157
	0	3Y	14.75	14.8277
	0	4Y	15.25	15.3195
	0	5Y	15.50	15.5669
	0	7Y	15.50	15.5681
	0	10Y	15.50	15.5754
	2	2Y	14.75	14.2608
	2	3Y	14.75	14.4294
	2	4Y	15.25	15.1731
	2	5Y	15.50	15.4474
	2	7Y	15.50	15.3841
	2	10Y	15.50	15.4990

With a lag of zero the term-sheet spreads are at most 0.08 basis points from the quotes; with a lag of two, up to 0.49. Two days of coupon timing matter much less than two days of notional timing, so the curve uses a lag of zero. The cost is under a tenth of a basis point, well inside a typical bid-offer spread.

The curve

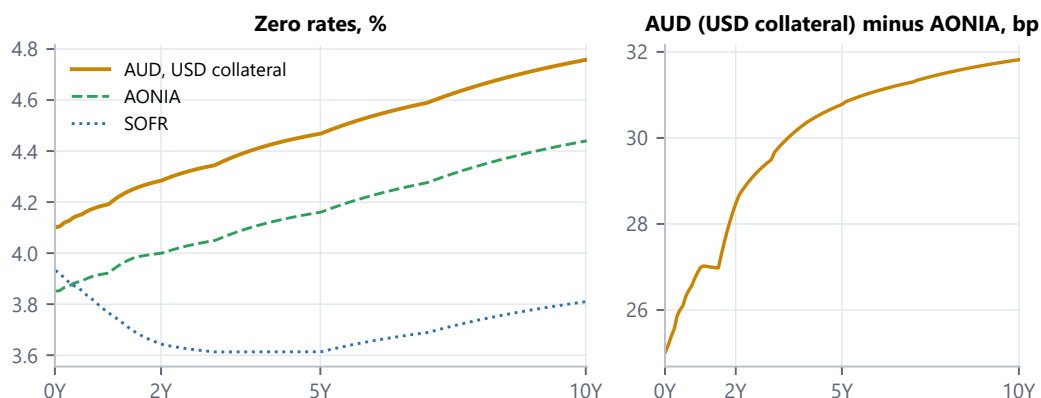


Figure 8.1. Zero rates (continuous, actual/365 fixed) of the Australian dollar curve under US dollar collateral, AONIA and SOFR, and the gap between the first two. Illustrative quotes.

The new curve sits above AONIA all the way out: about 27 basis points at one year, 31 at five and 32 at ten. At five years its zero rate is 4.47%, against 4.16% for AONIA.

Where the thirty basis points come from

Why about thirty? The basis is quoted against **BBSW**, not AONIA, and BBSW itself sits above AONIA by the AONIA/BBSW spread of *AUD Swaps & Curves*. So, roughly:

$$\text{gap to AONIA} \approx \underbrace{\text{cross-currency basis}}_{\text{BBSW/SOFR}} + \underbrace{\text{AONIA/BBSW spread}}_{\text{single currency}}$$

At five years that is $15.5 + 15.5 = 31$ basis points, against an actual gap of 30.8. The approximation ignores compounding and the different day counts, but it explains the level.

Summary

- **FX swaps** build the curve out to a year and **BBSW/SOFR basis swaps** take it to ten, one discount factor per quote.
- QuantLib's `MtMCrossCurrencyBasisSwapRateHelper` needs three flags: USD collateral, spread on the AUD leg, USD notional resetting.
- Every quote reprices, a par basis swap valued by hand is worth **zero**, and the payment-lag choice costs **under a tenth of a basis point**.
- The curve sits about **thirty basis points above AONIA**: the cross-currency basis plus the AONIA/BBSW spread.

Sources

Companion notebook: `s2e08_building_the_curve.ipynb`. Every number in this chapter is computed by it.

1. Australian Financial Markets Association, *Amended Proposed Term Sheet for BBSW/SOFR Cross Currency Basis Swaps (2024)*. https://afma.com.au/getattachment/Policy/topics/Benchmarks-Ibor-Transition/Sections/Content/AMENDED-Proposed-Term-Sheet_BBSW-SOFR-BASIS-SWAP_2024.pdf?lang=en-AU
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Pricing with the Curves

Two prices no single quote gives you: an FX forward five and ten years out, and a basis swap that is no longer at the market. And why the resetting US dollar leg is worth nothing.

Forwards to any date

FX swap quotes stop at one year, but the curve of Chapter 8 goes to ten. For any value date T , the outright forward is spot times the ratio of discount factors from spot:

$$F(T) = S \times \frac{P_{\text{AUD|USD}}(s, T)}{P_{\text{SOFR}}(s, T)}$$

For comparison, the table also shows the same formula with **AONIA** in place of the Australian dollar curve under US dollar collateral: the forward you would get if you ignored the cross-currency basis.

AUD/USD outright forwards from the curve, and with AONIA instead. Spot 0.7100. Illustrative quotes.

YEARS	VALUE DATE	FROM THE CURVE	POINTS	USING AONIA	DIFFERENCE (POINTS)
1	29 Sep 2027	0.7070	-30.3	0.7089	-19.1
2	29 Sep 2028	0.7009	-91.1	0.7049	-40.2
5	29 Sep 2031	0.6802	-297.9	0.6908	-105.6
10	29 Sep 2036	0.6457	-643.3	0.6666	-209.0

Five years out the forward is 0.6802, 298 points below spot. With AONIA you would get 0.6908: 106 points too high at five years, and 209 at ten. The error grows with maturity because the basis compounds: the discount factor ratio differs by roughly the gap in rates times the years.

What ignoring the basis costs

Price a five-year forward on AUD 100 million with AONIA, and the forward rate is wrong by the difference in the table. That error is paid in US dollars in five years, so its value today, in Australian dollars, is:

$$\text{Error} = N (F_{\text{AONIA}} - F) \frac{P_{\text{SOFR}}(T)}{F(\text{today})}$$

It comes to **AUD 1,240,926**. The basis looks small, a few tenths of a percent a year, but over five years on a trade this size it adds up to real money.

An off-market basis swap

Now a trade that has been on the books for a while. An Australian bank hedging its US dollar debt pays the Australian dollar leg of a five-year BBSW/SOFR basis swap, BBSW plus 12 basis points on AUD 100 million, and receives SOFR. Today's five-year basis is 15.5. So the bank pays 3.5 basis points a year less

than a new swap would cost. What is that worth?

Every cash flow. Value all 20 quarters of both legs with the loop of Chapter 6, on the curves of Chapter 8. The swap is worth AUD 156,338.77 to the bank.

The shortcut. A swap at today's basis is worth zero; that is what "today's basis" means. The traded swap differs from it only in the spread, so its value is the spread difference times the value of one basis point of spread on the Australian dollar leg:

$$V_{\text{bank}} = (s_{\text{market}} - s_{\text{traded}}) \times \text{BPV}, \quad \text{BPV} = \sum_i N \tau_i P_{\text{AUD|USD}}(p_i) \times 0.0001$$

Here the BPV is AUD 44,668, and the shortcut gives AUD 156,338.77.

The two valuations in the notebook.

```
market_bp = m.basis["5Y"].value() * 1e4
rec = mtm_swap(m, "5Y", traded_spread_bp / 1e4, notional_aud, coupon_lag=0) # value to the AUD-Leg receiver
value_bank = -rec["value"] # the bank pays the AUD Leg
bp_value = rec["bps"]
shortcut = (market_bp - traded_spread_bp) * bp_value
par_check = mtm_swap(m, "5Y", market_bp / 1e4, notional_aud, coupon_lag=0)["value"]
```

	VALUE TO THE BANK
Every cash flow	AUD 156,338.77
Spread gap × BPV	AUD 156,338.77
A swap at today's basis	AUD 0.00

The two agree to a fraction of a cent, and a swap at today's basis comes out at zero, as a market price should.

Where the value sits

Split the value by leg and something striking appears.

	VALUE TO THE BANK
Australian dollar leg (including the notional exchanges)	AUD 156,339
US dollar leg (including the resets)	AUD 0

The US dollar leg is worth **nothing**. The reason is the reset. Each quarter, the US dollar leg is a fresh loan: the notional NX_i changes hands at the start of the quarter and comes back at the end with SOFR interest. Discounted on SOFR, and with the coupon paid at the end of the quarter, that is worth

$$NX_i \left[P_{\text{SOFR}}(t_{i-1}) - P_{\text{SOFR}}(t_i) (1 + R_i \tau'_i) \right] = 0,$$

because compounded SOFR over the quarter is exactly $1 + R_i \tau'_i = P_{\text{SOFR}}(t_{i-1}) / P_{\text{SOFR}}(t_i)$ on the SOFR curve. A floating-rate loan at the discounting rate is worth par, every quarter. So all of the value sits in the Australian dollar leg, in that spread.

With the term sheet's two-day payment lag the bracket is not exactly zero, but very nearly: Chapter 8 measured the effect at under a tenth of a basis point.

Summary

- The curve gives an **FX forward for any date**: $F = S \times P_{\text{AUD|USD}} / P_{\text{SOFR}}$.
- Using AONIA instead ignores the basis and **misprices long forwards**, by more the longer the forward.
- An off-market basis swap is worth **the spread gap times the BPV** of the Australian dollar leg.
- The **resetting US dollar leg is worth zero**: each quarter it is a floating-rate loan at the discounting rate.

Sources

Companion notebook: `s2e09_pricing.ipynb`. Every number in this chapter is computed by it.

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Risk

A five-year basis swap and a five-year FX forward both swap Australian dollars for US dollars, but what they are exposed to could hardly be more different. Bump the quotes, rebuild every curve, revalue, and hedge.

Two trades

Both trades are on AUD 100 million for five years, valued on the curves of Chapter 8.

- **The basis swap** of Chapter 9. The bank pays the Australian dollar leg, BBSW plus a spread below today's basis, and receives SOFR on a resetting US dollar notional. It is worth AUD 156,339 to the bank.
- **An FX forward.** The bank buys AUD 100 million for US dollars for value on Monday 29 September 2031, at today's five-year outright, 0.6802. At that rate it is worth nothing today.

Bump and revalue

Every sensitivity in this chapter is measured the same way. Move a set of market quotes up, let every curve that depends on them rebuild, and value the trade. Move them down by the same amount and value it again. Half the difference is the **central difference**: the change in value per bump, accurate to second order.

Four risk factors:

FACTOR	BUMP
Basis	every BBSW/SOFR basis swap quote +1 bp
SOFR	every SOFR swap rate +1 bp and every SOFR future price -0.01 (its rate +1 bp)
AUD rates	every AONIA swap rate +1 bp; the BBSW curves are built on AONIA, so they move too
FX spot	AUD/USD +1%

In QuantLib this is easy because every quote is a `SimpleQuote` object that the rate helpers observe. Change a quote's value and every curve built on it recalculates the next time it is used.

Central differences over the four risk factors.

```
def central(fn, up, down):
    up(); a = fn(); down(); down(); b = fn(); up()
    return (a - b) / 2

sofr_rates = [q for k, q in m.sofr_q.items() if not k.startswith("SR3")]
sofr_futs = [q for k, q in m.sofr_q.items() if k.startswith("SR3")]
aonia_q = [q for (c, t), q in m.aud_q.items() if c == "AONIA"]
S0 = m.S.value()
factors = {
    "basis": (lambda: bump(list(m.basis.values()), 1e-4), lambda: bump(list(m.basis.values()), -1e-4)),
    "sofr": (lambda: (bump(sofr_rates, 1e-4), bump(sofr_futs, -0.01)),
            lambda: (bump(sofr_rates, -1e-4), bump(sofr_futs, 0.01))),
    "aud_rates": (lambda: bump(aonia_q, 1e-4), lambda: bump(aonia_q, -1e-4)),
    "fx": (lambda: m.set_spot(m.S.value() + 0.01 * S0), lambda: m.set_spot(m.S.value() - 0.01 * S0)),
}
labels = {"basis": "Basis +1 bp", "sofr": "SOFR +1 bp", "aud_rates": "AUD rates +1 bp", "fx": "AUD/USD +1%"}
```

What each bump does

Change in value today for each bump, from the bank's side. Illustrative quotes.

BUMP	BASIS SWAP (AUD)	FX FORWARD (AUD)
Basis +1 bp	+44,632	-39,893
SOFR +1 bp	-3	+39,428
AUD rates +1 bp	-35	-38,782
AUD/USD +1%	+1	+799,222

The basis swap is almost pure basis risk. It gains AUD 44,632 for each basis point the market basis rises, very close to the value of one basis point of spread in Chapter 9, and SOFR, Australian rates and the exchange rate barely touch it. Chapter 9 showed why: the resetting US dollar leg is worth about nothing at any rates and any exchange rate, and the Australian dollar leg's floating coupons move with the curves that discount them. What's left is the spread.

The forward is the opposite. A 1% rise in AUD/USD is worth AUD 799,222. And it carries interest rate risk in both currencies: it gains AUD 39,428 per basis point on SOFR and loses AUD 38,782 per basis point on Australian rates, and AUD 39,893 on the basis.

Why the forward moves

Write the forward's value using covered interest parity. With $F(T) = X_0 P_{\text{AUD|USD}}(T) / P_{\text{SOFR}}(T)$, where X_0 is today's rate:

$$V = N(F(T) - K) \frac{P_{\text{SOFR}}(T)}{X_0} = \underbrace{N P_{\text{AUD|USD}}(T)}_{\text{long an AUD zero-coupon bond}} - \underbrace{\frac{NK}{X_0} P_{\text{SOFR}}(T)}_{\text{short a USD zero-coupon bond}}$$

A long Australian dollar forward is a long five-year Australian dollar zero-coupon bond and a short US dollar one. Every row of the table follows:

- **SOFR up:** the US dollar bond the bank is short falls in value, and the bank gains.
- **Australian rates or the basis up:** $P_{\text{AUD|USD}}$ is built on BBSW plus the basis, so it falls, and the bank's long Australian bond loses.

- **AUD/USD up:** the US dollar bond, worth $NK P_{\text{SOFR}}(T)$ in US dollars, is worth fewer Australian dollars, so the position gains about 1% of the Australian bond's value, $N P_{\text{AUD|USD}}(T) = \text{AUD } 79,927,852$.

Each bond moves by roughly its value times its five years to maturity per basis point, and the two bonds are worth about the same, which is why the three rate sensitivities in the table are so close in size. SOFR and the Australian rates come with opposite signs: if both curves move together, most of the risk cancels. What the forward really holds, besides the exchange rate, is the **gap** between two interest rates.

Basis risk by tenor

Now bump the basis quotes one tenor at a time.

Change in value for +1 bp on one basis swap quote at a time. The totals equal the parallel bump.

BASIS QUOTE BUMPED	BASIS SWAP (AUD)	FX FORWARD (AUD)
2Y	-6	+731
3Y	-9	+1,151
4Y	-12	+1,615
5Y	+44,659	-43,390
7Y	0	0
10Y	0	0
Total	+44,632	-39,893

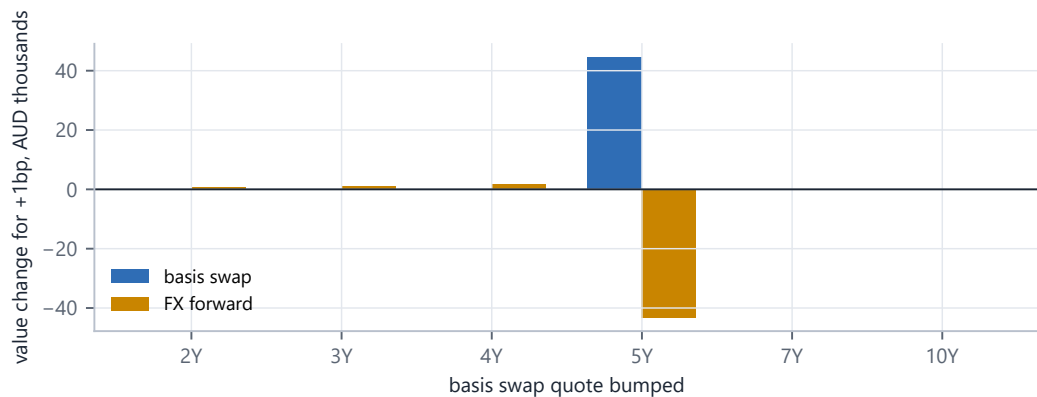


Figure 10.1. Basis risk by tenor for the two trades, in AUD thousands per basis point.

The basis swap's risk sits at five years, where it matures. The forward's is mostly there too, AUD 43,390, with small offsets at two, three and four years. Those come from the bootstrap: the five-year discount factor is solved *through* the shorter quotes, so if a shorter basis rises while the five-year quote stays put, the curve must bend back at five years to keep the five-year swap at par. Seven and ten years don't matter to either trade. The buckets add up to the parallel numbers, as they should: the totals row matches the first row of the previous table.

Hedging the forward

The bank wants to keep the forward but not its basis or FX risk. It adds two trades.

For the basis, it pays the Australian dollar leg of a new five-year basis swap at today's basis. Paying that leg gains when the basis rises, just as the forward loses. It is sized so the two cancel: AUD 89.3 million of notional, found by dividing the forward's basis risk by one basis point of spread per Australian dollar of notional.

For FX, it sells Australian dollars spot. A spot sale of Y loses about $0.01 Y$ when AUD/USD rises 1%, so the bank sells AUD 80.0 million: not the full hundred million, but about the forward's Australian dollars discounted to today, AUD 79.9 million. It is the present value that moves with spot.

The forward's risk before and after the two hedges.

BUMP	FORWARD ALONE (AUD)	WITH BOTH HEDGES (AUD)
Basis +1 bp	-39,893	0
SOFR +1 bp	+39,428	+39,428
AUD rates +1 bp	-38,782	-38,782
AUD/USD +1%	+799,222	0

Basis and FX risk are gone. What is left is the rate risk in the two currencies, large but opposite, which a desk would hedge with SOFR and AONIA swaps, the instruments of Chapter 3 and *AUD Swaps & Curves*.

One swap can only hedge one number. The five-year basis swap cancels the forward's **parallel** basis risk, but not its tenor-by-tenor profile:

Basis risk by tenor after the parallel hedge.

BASIS QUOTE BUMPED	FORWARD + HEDGES (AUD)
2Y	+731
3Y	+1,151
4Y	+1,615
5Y	-3,497
7Y	0
10Y	0

The pieces now add up to zero but don't vanish; the largest is AUD 3,497. A desk that wanted them gone too would add small two-, three- and four-year basis swaps, one per bucket.

Where the series ends

This book started from a single overnight rate published by the New York Fed. It compounded SOFR by hand, built the SOFR curve, read AUD/USD spot and forwards, met the cross-currency basis in FX swaps and basis swaps, saw why collateral decides discounting, and bootstrapped an Australian dollar curve for US dollar collateral. This last chapter turned that curve into prices, risks and hedges. The same few ideas carry every step: a quote pins down a discount factor, a trade is a sum of discounted cash flows, and every number can be checked by hand.

Summary

- Risk here is **bump and revalue**: move the quotes, rebuild every curve, halve the up-and-down difference.
- A resetting **basis swap is almost pure basis risk**, concentrated at its maturity.
- An **FX forward** is a long zero-coupon bond in one currency and a short one in the other: **FX, both rate curves and the basis**.
- Tenor **buckets add up to the parallel bump**.
- Hedge **basis with a basis swap** and **FX with spot**, sized to the present value; the leftover rate risk goes to SOFR and AONIA swaps.

Sources

Companion notebook: `s2e10_risk.ipynb`. Every number in this chapter is computed by it.

1. Federal Reserve Bank of New York, *Terms of Use (Reference Rates use restrictions)*. <https://www.newyorkfed.org/privacy/termsfuse>
2. Reserve Bank of Australia, *The Financial Crisis through the Lens of Foreign Exchange Swap Markets (Bulletin, June 2010)*. <https://www.rba.gov.au/publications/bulletin/2010/jun/7.html>

Appendix: USD, FX and cross-currency conventions at a glance

These are the conventions used throughout the book, with the source for each. "AFMA" means AFMA's *Interest Rate Derivative Conventions* (June 2025) unless stated otherwise, and "term sheet" AFMA's *Amended Proposed Term Sheet for BBSW/SOFR Cross Currency Basis Swaps* (2024). Always check the current documents and your own documentation before relying on them.

US dollar rates

RATE	CONVENTION	SOURCE
Federal funds target range	set by the FOMC, 25 bp wide; held in place by interest on reserve balances (IORB) and the overnight reverse repo (ON RRP) rate	Federal Reserve Board, <i>Implementation Note; IORB FAQs; ON RRP Operations</i>
EFFR	volume-weighted median of overnight federal funds trades; published about 9:00am New York time the next business day	New York Fed, <i>Additional Information about Reference Rates</i>
SOFR	volume-weighted median of overnight Treasury repo (tri-party, GCF and FICC-cleared bilateral); published about 8:00am New York time the next business day; US government securities market calendar	New York Fed, <i>Secured Overnight Financing Rate Data; Additional Information about Reference Rates</i>
SOFR Index	SOFR compounded on every business day since 2 April 2018 (starting at 1.00000000), eight decimal places, published on its value date	New York Fed, <i>Additional Information about Reference Rates</i>
Money-market day count	actual/360	ARRC, <i>A User's Guide to SOFR</i>

SOFR overnight index swaps

ITEM	CONVENTION	SOURCE
Start	two New York business days after the trade (T+2)	CFTC, <i>Swaps Made Available to Trade</i>
Day count	actual/360, both legs	CFTC, <i>Swaps Made Available to Trade</i>
Payments	annual, both legs; one payment for a year or less	CFTC, <i>Swaps Made Available to Trade</i>
Payment lag	two business days after each period ends	CFTC, <i>Swaps Made Available to Trade</i>
Floating amount	$N[\prod(1 + r_i n_i/360) - 1]$, one fixing per government securities business day	ARRC, <i>User's Guide to SOFR</i>

Three-month SOFR futures

ITEM	CONVENTION	SOURCE
Reference quarter	third Wednesday of the contract month to the third Wednesday three months later (IMM to IMM)	CME Submission 18-069 to the CFTC
Final settlement	price = 100 – SOFR compounded over the reference quarter	CME Submission 18-069
Contract size	USD 25 per basis point	CME Submission 18-069
Convexity adjustment	set to zero in this book (a simplification)	—

AUD/USD foreign exchange

ITEM	CONVENTION	SOURCE
Quote	US dollars per Australian dollar, four decimal places	Federal Reserve Board, H.10
Spot	two good business days in both Sydney and New York (T+2)	AFMA §3.6; BIS Triennial Survey definitions
Forward points	forward minus spot, in units of 0.0001	QuantLib, FxSwapRateHelper documentation
Overnight, tom-next	FX swaps from today to tomorrow, and from tomorrow to spot	QuantLib, FxSwapRateHelper documentation
Covered interest parity	$F = S \times P_{\text{AUD}}(s, T) / P_{\text{USD}}(s, T)$	RBA <i>Bulletin</i> , June 2010

BBSW/SOFR cross-currency basis swaps

ITEM	CONVENTION	SOURCE
AUD leg	3-month BBSW + spread on a constant AUD notional; actual/365 fixed	term sheet
USD leg	SOFR compounded daily, no spread, actual/360; USD notional reset each quarter at the FX rate	term sheet
Payments	quarterly, two business days after each period ends	term sheet
Notionals	exchanged at the start and the end; reset amounts on the roll dates	term sheet
Start	T+2 on Sydney and New York business days	term sheet; AFMA §3.6
Quote	spread on the AUD leg, priced on US dollar collateral discounted at SOFR	AFMA §3.6; term sheet
Clearing	not centrally cleared	AFMA §2.3

Discounting

TRADE	DISCOUNT CURVE	SOURCE
AUD swaps, cleared or not	AONIA	AFMA §3.6
USD swaps cleared at LCH	SOFR	LCH self-certification to the CFTC (2020)
AUD cash flows under USD collateral	$P_{\text{AUD USD}}(T) = F(T) P_{\text{SOFR}}(T) / F(\text{today})$	Fujii, Shimada and Takahashi (2009)

Glossary

Basis (cross-currency). The spread on the Australian dollar leg of a basis swap that makes it fair; equivalently, the extra cost of borrowing a currency through FX swaps rather than directly.

BPV. The value of one basis point of spread (or rate) on a swap leg: $\sum_i N \tau_i P(p_i) \times 0.0001$.

Central difference. A sensitivity measured by bumping up and down and halving the difference in value.

Collateral rate. The interest paid on posted collateral, usually the overnight rate of its currency; a fully collateralised cash flow is discounted at it.

FX swap. Two exchanges of currencies at rates agreed today: one on a near date, the reverse on a far date.

IMM date. The third Wednesday of March, June, September or December (and, for SOFR futures, of any contract month).

Mark-to-market (resetting) swap. A cross-currency swap whose notional in one currency resets each period to the FX rate of the day, with the difference paid over.

Outright forward. An exchange of currencies on a date after spot, at a rate agreed today.

Point (pip). 0.0001 of the exchange rate; forward points are quoted in these units.

Spot date. The settlement date of a spot deal: two good business days in both currencies' financial centres.

Zero-coupon bond. A single payment at maturity; a long FX forward is a long zero-coupon bond in one currency and a short one in the other.